Geophysical Studies of North African Cenozoic Volcanic Areas: III. Garian, Libya

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Paleomagnetic and K/Ar whole rock absolute age data are described for material from the Garian area of Libya, centered at 13°E, 32°N. Within-unit cleaned paleomagnetic directions from the essentially unaltered lavas are very well defined and can almost certainly be taken as reliable measurements of the geomagnetic field direction during the initial cooling of each flow. However, the distributions of mean direction, from which the effect of repeated sampling of the field at one time has been removed, does not suggest that a reversing axial dipole field has been recorded in a representative manner. Both N and R groups of directions are azimuthally elongated, and the average poles for the N and R groups differ by 21°, or four times the 95% level uncertainty for each average pole. A number of possible physical explanations for the paleomagnetic results are discussed. The conventional overall average pole at 88°N, 123°E, δ_p: 3°, δ_m: 7° does not differ significantly from the geographic pole, a result which agrees closely with that of Schult and Soffel (1973). However, the value of these overall average poles in estimating absolute plate motion must await an understanding of the sources of the asymmetries in the paleomagnetic data.

Cet article traite des données paléomagnétiques et des âges K/Ar sur roche totale obtenus à partir d'échantillons provenant de la région de Garyan en Lybie. Le centre de cette région possède les coordonnées 13°E et 32°N. Les directions paléomagnétiques recueillies dans chaque unité de lave fraîche sont très bien définies et représentent des mesures à peu près fiables de la direction du champ géomagnétique lors du début du refroidissement de chaque coulée. Toutefois, les distributions de la direction moyenne, dont l'effet d'échantillonnage répété du champ a été annulé, ne suggèrent pas un enregistrement représentatif du renversement de l'axe d'un champ dipolaire. Les groupes de directions N et R sont allongés dans l'azimut et leurs pôles moyens diffèrent par 21°, soit, pour chaque pôle, quatre fois le niveau de confiance de 95 pourcent. Un certain nombre d'explications physiques sont discutées. Le pôle moyen conventionnel à 88°N et 123°E, δ_p: 3° et δ6m: 7°, n'est pas éloigné significativement du pôle géographique un résultat en accord avec celui de Schult et Soffel (1973). L'utilisation de ces pôles moyens dans l'estimation des mouvements absolus des plaques doit cependant attendre que nous comprenions les sources des asymétries des données paléomagnétiques. [Traduit par le journa]

Introduction

This is the third paper in a series describing combined paleomagnetic and K/Ar whole rock absolute age studies of Neogene and younger volcanic areas of stable North Africa. (Ade-Hall *et al.* 1974, I; Ade-Hall *et al.* 1975, II). A fourth paper (in preparation) will discuss the collected evidence and will give new K/Ar ages for the Tibesti and Eghei volcanics. The purpose of the studies is to supply precise data for evaluation of the absolute motion of the African Plate during the Neogene. The chain of volcanic areas studied, from near to the Mediterranean littoral in Libya to Tibesti in Chad are fairly linearly distributed and may mark a plume trace (Fig. 1). Clearly it is worth testing this chain for this type of origin, and, in order to make the evidence available with the least delay, the complete results for each area are being published separately.

In this paper results from the most northerly area, the basaltic and phonolitic volcanic area close to Garian in Libya, approximately 100 km south of Tripoli, are described.

The volcanic rocks of the Garian area cover some 3000–4000 km^2 centered on 32°N, 13°30′E. Two distinct rock types and modes of occurrence of igneous material are known. Alkali plateau basalts occur only as a capping to the Ceno-

TABLE 1. Location of sampling sites in the Garian area. Aerial photograph numbers refer to field locations marked on sheets of the 1:24 000 Tripolitania (Wadi Meginin Catchment Area) series, Fairey Surveys Ltd., Reform Rd., Maidenhead, Berkshire, U.K.

Site	Aerial Photograph	Location
Lava Section containing flows A10C001-18	Sheet 13	2.5 km SSE of As Salihat village at 13°7.2′E, 32°9.8′N
Lava Section containing flows A10C021-31	17	13°15′E, 32°7′N
Lava Section containing flows A10C032-38	16	13°6′E, 32°4′N
Phonolite annular intrusion A10C020	13	13°2′E, 32°10′N
Phonolite intrusion A10C040	13	13°1′E, 32°15′N
Phonolite annular intrusion A10C041	13	13°1′E, 32°15′N
Phonolite intrusion A10C042	13	12°59′E, 32°15′N
Phonolite intrusion A10C043	13	12°59′E, 32°16′N

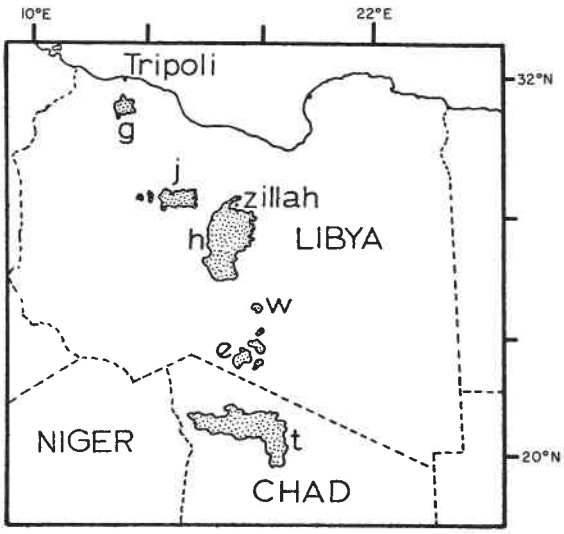


FIG. 1. The geographic distribution of the Cenozoic volcanic areas described in this series of papers.
g: Garian Volcanic Area (this paper, III)
j: Jebel Soda (Paper II)
h: Haruj Assuad (Paper I)
w: Waw en Namus
e: Eghei (Paper IV)
t: Tibesti (Paper IV).

manian – Turonian sediments of the upper part of the Jebel Nefusa escarpment. Phonolite plugs, domes and ring dikes outcrop both on the top of the Jebel Nefusa escarpment and in the upper Triassic and lower Jurassic limestones at the base of the escarpment. Although these alkali basalts and phonolites are closely related spatially, and are separated from the next area of post-Cretaceous igneous activity by 400 km, there is considerable doubt as to whether they belong to the same period of activity. Aside from obvious chemical differences, the basalts have been considered on general appearance to be considerably younger than the phonolites. Volcanic landforms are not preserved in the basalt area, as they are in the Pliocene and younger Haruj Assuad basalts further south (Paper I of this series). However, this may be in part the result of more rapid weathering in an area of relatively high rainfall. The basalts of the Garian area are generally extremely fresh, with empty vesicles, in contrast to the common occurrence of zeolite infilling in older basalts, for example, the Upper Miocene Jebel Soda basalts (Paper II of this series). Radiometric ages conflict with the rather weak geological evidence for the age of the volcanics. Piccoli (1970) gives mean ages of 39.4 m.y. for the phonolites and 42 $\pm$ 10 m.y. (range from 29 to 53.5 m.y.) for some of the basalts, while others are dated at 3.5–5.7 m.y. The dates reported in this paper are close to the lower ages obtained by Piccoli although they are for flows in the area yielding his older ages. These differences are discussed later. Previous paleomagnetic results for the area are reported in Schult and Soffel (1973).

Sampling

Thirty-six basaltic flows comprising three short sections, together with five phonolite bodies, were sampled. Each volcanic unit was represented by four geographically or solar oriented drill cores, sampling being carried out by portable gasoline powered drills of the U.S.G.S. type. The location of the lava flow sections and phonolite bodies is given in Table 1.

Natural Remanent Magnetism of the Garian Basalts and Phonolites

The natural remanent magnetization of the material was measured using a Schonstedt

TABLE 2. Garian — basaltic lavas and phonolite intrusions, temporally independent paleomagnetic directions and pole positions

	Unit numbers	D(°)	I(°)	k	α_{95}(°)	λ(°N)	ϕ(°E)	δ_p(°)	δ_m(°)	n used/n collected	
Basalt Section (C001 top flow)	A10C001/2	012	+62	536	3.5	75	051	4.2	5.4	—	009, +64; 010, +59
	C003	018	+62	378	2.6	72	058	3.1	4.0	4/4	
	C004	008	+51	1124	1.5	83	100	1.4	2.0	4/4	
	C005	012	+62	418	2.4	76	051	2.9	3.7	4/4	
	C006	351	+66	726	2.6	72	354	3.5	4.3	2/4	
	C007	009	+56	210	3.4	81	066	3.6	4.9	4/4	
	C008/9	014	+71	1120	3.0	64	032	4.5	5.2	—	009, +71; 020, +71
	C010/11	006	+56	9601	1.0	83	056	1.1	1.5	—	005, +56; 007, +56
	C012/14	006	+62	2199	1.7	78	036	2.1	2.7	—	004, +62; 005, +61; 010, +62
	C015	Scattered directions, probable downwards inclination. transition direction?									
	C016	158	−40	112	5.4	69	263	3.9	6.5	3/4	
	C017	172	−45	422	2.4	81	247	1.9	3.0	4/4	
	C018	169	−36	509	2.2	74	237	1.5	2.6	4/4	
Basalt Section 2 (C021 top flow)											
	C021	007	+44	740	1.8	81	144	1.4	2.3	4/4	
	C022	015	+45	913	1.6	76	122	1.3	2.1	4/4	
	C023	015	+34	151	4.7	71	144	3.0	5.3	3/4	
	C024	012	+47	1101	1.5	79	121	1.2	1.9	4/4	
	C025/6	015	+41	1765	2.4	74	132	1.7	2.9	—	014, +41; 017, +40
	C027	058	−01	13	13.9	26	122	7.0	13.9	4/4	Intermediate direction, not included in average directions
	C028/9	185	−57	2859	1.9	83	049	1.9	2.7	—	187, −57; 184, −56
	C030	223	−26	45	8.6	47	118	5.0	9.2	3/4	Direction not included in averages
	C031	167	−45	752	1.8	78	263	1.5	2.3	4/4	

TABLE 2. (*Concluded*)

	Unit numbers	D(°)	I(°)	k	α_{95}(°)	λ(°N)	ϕ(°E)	δ_p(°)	δ_m(°)	n used/n collected	
Basalt Section 3 (C032 top flow)											
	C032	169	−28	70	5.9	70	228	3.6	6.5	4/4	
	C033/4	173	−37	2911	1.8	77	225	1.3	2.2	—	173, −38; 173, −36
	C035	005	+53	459	2.3	85	078	2.2	3.2	4/4	
	C036/8	163	−45	2579	1.6	75	268	1.3	2.0	—	164, −44; 165, −45 161, −46
Phonolite Units											
	C020	No consistent direction									
	C040	No consistent direction									
	C041	No consistent direction									
	C042	No consistent direction									
	C043	182	−46	207	3.4	85	168	2.8	4.4	4/4	Not included in basalt average
Mean direction and pole for 15 normal polarity basalt temporally independent directions		010	+54	55	4.9	81	080	3.2	6.9	$n = 15$	
Mean direction and pole for 8 reverse polarity basalt temporally independent directions		169	−42	64	6.2	77	249	3.6	7.6	$n = 8$	
Mean direction and pole for all 23 basaltic temporally independent directions		002	+50	35	5.0	88	123	3.2	6.7	$n = 23$	
Mean pole for all 23 basaltic temporally independent directions		—	—	—	—	88	125	$K = 29.9$, $A_{95} = 5.30$ on poles			

SSM-1a spinner magnetometer. Systematic alternating field remanence demagnetization was carried out on all specimens. For the basalts, where in all cases the stable component dominated the NRM, the resulting unit mean directions of cleaned remanent magnetization are well defined, with only six α_{95} values exceeding 4°. Fisher statistical parameters have been used to provide a familiar measure for the small within-flow scatter of directions (Table 2). However, it should be kept in mind that a set of four directions cannot be expected to have a true Fisherian distribution. One flow (C015), probably magnetized during polarity transition, failed to yield a satisfactory unit mean direction. In contrast, the phonolite specimens generally had soft remanent moments that frequently failed to give stable end-directions. Only one phonolite unit (C043) yielded a satisfactory unit mean direction. It is not possible even to quote polarities for the other four units. All three basalt sections record polarity reversals, with one reversal recorded in two sections and two in the third section. For the purposes of attempting to estimate absolute plate motion the directions for the individual flows were reduced to a restricted number of temporally independent directions before computation of an overall average direction was contemplated. This operation, which is described in detail in Paper I, was designed to remove any bias in the paleomagnetic data resulting from repetitive sampling of the geomagnetic field at one point in space and time. Application of this operation reduced the number of paleomagnetic data available for an overall average from 35 to 25 (Table 2 and Fig. 2).

Close clustering within the N and R polarity groups is evident in Fig. 2, as is a tendency in both groups for azimuthal elongation of distributions. Another feature of interest is the difference in the mean directions and poles for the N and R distributions. The mean poles for these distributions, at 81°N, 080°E (N) and 77°N, 249°E (R) (Table 2) are separated by 21°, or approximately four times the 95% confidence limits on each pole. It seems that, having gone to some trouble to remove a considerable degree of duplication in the data arising from eruptive irregularity, the reduced data do not have the appearance of a representative sample of an axial dipole field of alternating N and R polarity. Several explanations for the various asymmetries in the direction distributions can be envisioned.

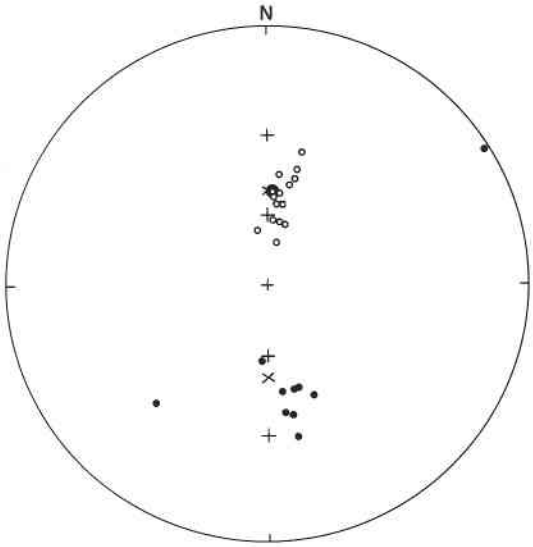


FIG. 2. Stereoplot of the temporally independent paleomagnetic directions for the Garian lavas. Open circle: lower hemisphere direction. Filled circle: upper hemisphere. Large double circle: overall average direction. Crosses: centered axial dipole directions for the present latitude of the sampling area.

Lack of exact opposition of N and R groups of cleaned directions has been observed in several lava collections, notably from the Keweenawan and Bermuda sequences; the latter also shows a degree of streaking of directions (Palmer 1970; Ade-Hall *et al.* 1973). It seems likely that the magnetization of the Keweenawan and Bermuda sequences, both of which have experienced epidote zone zeolite facies alteration, will be explained in terms of partial chemical remagnetization (CRM) associated with the observed extensive alteration of the magnetic minerals. However, unless such remagnetization can take place at close to ambient conditions, it seems unlikely that CRM acquisition can explain the direction distribution asymmetries of the essentially quite unaltered Garian lavas. The alternative explanation is imperfect recording of the geomagnetic field. Here we suppose that both azimuthal distributions and disagreement between N and R averages poles is a function of the relationship between the temporal sequence for the lavas and the geomagnetic field behavior over the whole eruptive interval. Two extreme circumstances can be envisioned. In the first the field is that of an axial dipole with secular variation averaging to zero over say 10^4 to 10^5 years. In

this case the asymmetrical distributions are explained by eruption occurring over two fairly short intervals, in periods of different polarity, each being significantly less than the time for secular variation to average to zero. Alternatively, eruption took place at approximately regular intervals during a period long enough for several reversals to have occurred but insufficient for secular or other variations to average to zero. The R–N–R sequence of asymmetrically disposed flows of section 3 (Table 2) suggests that the asymmetry can reasonably be accounted for by the presence of a nondipole field anomaly over the area, perhaps of crustal or upper mantle origin, of sufficient duration for several geomagnetic reversals to take place. It is of interest that Schult and Soffel (1973) who sampled at different sites in the Garian area from those sampled by the authors of this paper, find similar asymmetries in their results; further discussion and analysis of these asymmetrical distributions, together with similar features observed for the Haruj Assuad and Jebel Soda volcanic areas (Papers I and II of this series) will be given in Paper IV.

If the geomagnetic field during the eruptive cycle was dipolar with randomly distributed secular variation superimposed, a better estimate of the dipole field direction will be obtained by averaging the N and R directions. However, since this explanation for the data assumes the model involving incomplete sampling of the secular variation, the degree of improvement in the dipole field estimate obtained by averaging the N and R results cannot be ascertained. With flows C027 and C030 omitted, the overall average direction is 002, +50, with a corresponding pole at 88°N, 123°E, $\delta_p = 3°$, δ_m: 7°. The direction for flow C027 was omitted as its aberrant direction and situation between N and R sequences suggest it represents magnetization during a polarity transition. The direction for C030 was omitted on the criterion of Wilson *et al.* (1972), who suggest that the stable N and R states of the field are limited by equivalent pole positions within 40° of the geographic pole.

This overall average pole is not significantly different from the geographic pole. Schult and Soffel (1973), with different sampling sites and a different approach to data selection and reduction, obtain a pole at 86°N, 152°E, δ_p: 3°, δ_m: 4°. This does not differ significantly from our average pole and only differs by 1° at the 95% level from the geographic pole. We consider that the average virtual geomagnetic pole, probably involving *all* the information available from the Garian lavas, is now well known.

Conventionally, the close agreement between average paleomagnetic and geographic poles for the upper Miocene–Pliocene Garian lavas would be taken as implying at the most only very small latitudinal motion of this part of the African Plate over the time interval since volcanic activity took place. However, if the close agreement of the average pole, based on N and R average poles separated by 21°, and the geographic pole, is coincidental, which cannot be judged from the conventional data analysis given here, then absolute plate motion may have been larger than the apparent value of about 2°.

The temporally independent paleomagnetic directions can also be used to estimate the apparent paleosecular variation recorded in the Garian lavas. Following Doell (1970) and Cox (1969) the paleoangular standard deviation is found to be $17.1°^{+3.5°}_{-3.2°}$. This value is higher than, but not significantly different from, those predicted for the site latitude by several secular variation models. It should be kept in mind that, should it be possible in the future to show that the direction distributions for the Garian lavas are not simply the result of secular variation, then a different form of analysis for secular variation will be required.

Only one of five phonolite units yielded a well defined unit mean cleaned remanence direction. This direction, at N182°E, −46°, α_{95}: 3.4°, is not significantly different from the overall average direction for the basaltic flows. Schult and Soffel (1973) report directions for three phonolite bodies. All are reversed in sense and have associated α_{95} values of 15.5° to 23.8°. These large uncertainties preclude quantitative comparison with other paleomagnetic data from Garian igneous rocks.

Absolute Ages for Garian Lava Flows

Following careful examination of a number of thin sections of Garian lavas we selected four samples for dating by the K/Ar whole rock method. The analytical methods employed have been outlined in Paper I of this series.

The apparent ages range from 2.1 m.y. to 6.1 m.y. and the average is 3.8 m.y. (Table 3). These dates suggest that basaltic vulcanicity in this part of the Garian lava area occurred in upper

TABLE 3. K/Ar whole rock ages of Garian basalts

Unit	Core	^{40}Ar radiogenic (scc/g)	^{40}Ar radiogenic / ^{40}Ar total	$\%K_2O$	Apparent age (m.y.)
C014	1	2.6×10^{-8}	7	0.37	2.1 ± 0.8 (Uncertainty is two
C015	1	4.9×10^{-8}	27.5	0.44	3.4 ± 0.5 standard deviations)
C016	1	9.5×10^{-8}	71	0.47	6.1 ± 0.4
C034	1	3.8×10^{-8}	25	0.32	3.6 ± 0.4

Constants used: λ_e: $0.585 \times 10^{-10}y^{-1}$ $\lambda\beta$: $4.72 \times 10^{-10}y^{-1}$ $^{40}K/K$: 1.19×10^{-4} mole/mole

Miocene and Pliocene times, and that the duration of the volcanic activity responsible for the sampled flows was close to 4 m.y. These ages, representing units from two of the three lava sections sampled, have relatively large associated errors and thus cannot be tested against the established time-polarity scale of Cox (1969). It is possible to question the apparent range of ages for successive flows C014/015/016. While the order of the ages corresponds with the stratigraphic order of the flows, the apparent time span of about 4.0 m.y. may be excessive in view of the relatively high extrusion rate implied by the presence of one flow, C015, that appears to have recorded the geomagnetic field during a polarity transition.

Piccoli (1970) has reported ages of 37.9, 39.7 and 40.7 m.y. for the Garian phonolites. We present no additional absolute age data for the phonolites but suggest that the ages obtained by Piccoli should be reconsidered in view of their geographic proximity to the basalts and the lack of a significant difference between the remanence direction for phonolite C043 and the mean direction for the basalts. Possibly excess argon may be responsible for the high apparent ages of the phonolites. An $^{40}Ar/^{39}Ar$ stepwise heating study would allow this possibility to be tested. Piccoli also reports ages of 29, 37, 38, 52 and 53.5 m.y. for basalts from the area from which the samples described in this paper were obtained, and 3.5–5.7 m.y. for material from the southeastern part of the lava area. The apparent occurrence of basalts of such disparate ages in one volcanic area is unusual and is not found, for example, in Hawaii, Iceland or the Tertiary province of N.W. Britain where in each case duration of activity is always in the 1–10 m.y. range. In addition, the ages for the *northwestern* part of the lava area reported in this paper are similar to those found by Piccoli for the *southeastern* part of the area. Also, the low alteration state of the sampled lavas is consistent with a relatively young age. In conclusion, an age range for the Garian lavas of between 2 and 6 m.y. seems most likely. The age of the phonolites remains uncertain.

Summary

A combined paleomagnetic and absolute age study of the Garian lavas and phonolites of northwestern Libya has yielded a well defined pole position and satisfactory absolute ages for the lavas in addition to a single paleomagnetic direction for one of the five phonolite units sampled. For the lavas the pole position is at 88°N, 123°E with precision parameters of δ_p: 3° and δ_m: 7°. This pole is not significantly different from the geographic pole. Absolute latitudinal motion of the African Plate in the Garian area since the eruption of the lavas 4 ± 2 m.y. age may not have been more than 2°. The paleomagnetic results reported are in close agreement with those of Schult and Soffel (1973). A major new result reported here is convincing evidence for an upper Miocene–Pliocene age for the Garian lavas, in preference to a combination of Oligocene–Eocene and Pliocene ages as suggested by Piccoli (1970).

Acknowledgments

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